# Product pitch

Andre van Heerden

Open Window, Department of Fundamentals

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## 1. Project Title & Concept

**Application Name :** Tower of Arithmetic

**Tagline :**

**Category :** Education & EdTech (Gamified Learning)

**Short Overview :** Tower of Arithmetic is an AI-driven, turn-based RPG that transforms K–11 math practice into an immersive fantasy adventure. Players step into the shoes of a novice wizard apprentice forced to ascend a multi-floor trial tower designed by their archmage mentor. To cast elemental spells, disarm dungeon traps, and execute split-second defensive Quick-Time Events (QTEs), players solve dynamic math equations generated in real time by an integrated AI agent. When players make mistakes, the app provides real-time Socratic AI hints rather than flat penalties, reinforcing core concepts while keeping gameplay fluid.

## 2. Problem Statement & Research

### What is the Problem?

For millions of primary and secondary school students, practicing mathematics is experienced as a grueling, repetitive grind of static worksheets and high-stress drills. This traditional approach triggers widespread Math Anxiety—a psychological state of apprehension, tension, and fear that actively interferes with mathematical performance and cognitive processing (Ashcraft, 2002). Rather than engaging with mathematical concepts as creative tools for problem-solving, students view practice as an unpleasant chore to be avoided.

### Why is it Important?

Mathematics mastery is a foundational predictor of long-term academic success, financial literacy, and career readiness in STEM fields. When students develop early math avoidance,

it cripples their working memory capacity during problem-solving tasks, creating a self-fulfilling cycle of academic failure and low self-efficacy (Maloney & Beilock, 2012). Addressing this issue during practice is critical to maintaining student engagement in technical learning pathways.

## Who Experiences This Problem?

- **Primary & Secondary Students (Grades 1–11):** Suffer from test anxiety, cognitive overload during homework, and a complete lack of intrinsic motivation when practicing math.
- **Educators & Parents:** Struggle to keep unmotivated learners engaged using conventional homework tools and lack granular visual insight into why a student is struggling (e.g., conceptual misunderstanding vs. performance anxiety under timed pressure).

## Evidence & Research Base

- **High Prevalence of Math Anxiety:** Global PISA assessment findings reveal that over **39% of 15-year-old students** report feeling helpless or severely anxious when dealing with mathematics problems (OECD, 2023).
- **Working Memory Disruption:** Empirical research demonstrates that math anxiety temporarily degrades a student's working memory, making standard arithmetic feel significantly harder than it actually is (Ashcraft, 2002).
- **Flawed Educational Game Design:** Educational design literature highlights the failure of "chocolate-covered broccoli"—the practice of wrapping boring math drills inside unrelated mini-games. Studies show that educational games are only effective when learning mechanics are **intrinsically integrated** into core gameplay, rather than acting as artificial progress gates (Habgood & Ainsworth, 2011).

## Current Limitations & Gaps in Existing Tech

**1. Extrinsic Gameplay Integration ("Chocolate-Covered Broccoli")**

- **Competitor Example:** *Prodigy Math*
- **The Gap:** *Prodigy* wraps math in an engaging wizarding world, but completely detaches the learning from the gameplay loop. Players explore a fantasy map, encounter a monster, and then **the game abruptly pauses** to present a static pop-up quiz card. The math acts as a tollgate rather than a game mechanic.
- **Why it Fails:** Pausing gameplay to force a quiz card reinforces the perception that math is an unpleasant penalty required to access "the fun part," breaking player flow and triggering avoidance behaviors.

**2. Static Problem Banks & Limited Replayability**

- **Competitors Examples:** *Mathletics*, *Monster Math*, *Kahoot!*
- **The Gap:** These platforms rely on pre-scripted, hardcoded question pools and fixed template banks.
- **Why it Fails:** Students quickly recognize recurring problem structures, leading to answer memorization rather than deep conceptual understanding. Once a student completes a grade-level question bank, content runs dry, creating finite replay value and quick user churn.

**3. Lack of Adaptive Socratic Support & Punitive Error Handling**

- **Competitor Example:** *IXL Math*
- **The Gap:** *IXL* provides extensive skill drills, but when a student answers incorrectly, the app immediately slashes their progress score ("SmartScore") and dumps a static wall of text showing the full worked solution *after* the fact.
- **Why it Fails:** Punishing mistakes with score drops increases test anxiety and frustration (Ashcraft, 2002). Furthermore, presenting a pre-written post-mortem

answer fails to guide the student through the step-by-step logical breakdown mid-problem, leaving them helpless on the next attempt.

## Why *Tower of Arithmetic* Solves This

- **Intrinsic Magic Integration (vs. *Prodigy*):** Math is not a pause menu or a progress gate; **equations *are* the magic system** (Habgood & Ainsworth, 2011). Solving an equation directly calculates spell damage, while quick-time mental arithmetic forms real-time defensive shields.
- **AI-Driven Infinite Procedural Content (vs. *Mathletics* / *Monster Math*):** Powered by lightweight LLMs (Groq / Llama 3), equations and dungeon encounters are generated infinitely on the fly, tailoring difficulty dynamically across Grades 1–11 without repeating content.
- **Interactive Socratic AI Remediation (vs. *IXL*):** When a student makes a mistake during combat, an AI mentor steps in *mid-battle* to provide Socratic, hint-by-hint guidance rather than giving away the answer or penalizing points. This builds genuine conceptual mastery while keeping anxiety low.

### 3. Target Audience

## Primary Audience: Students (Grades 1–11 / Ages 6–17)

- **Who They Are:** K–11 learners experiencing math anxiety, homework burnout, or academic boredom with conventional learning tools.
- **Core Needs:** A low-friction, supportive practice environment that rewards effort, eliminates fear of failure, and builds mental math speed without high-stress penalties.

- **Frustrations:** Repetitive paper worksheets, rigid grading systems that penalize mistakes without explanation, and educational software that feels like disguised quizzes.
- **Why They Use It:** To experience math as an interactive magic system where solving equations unlocks elemental spells, disarms dungeon traps, and drives their avatar's progression through a fantasy world.
- **How It Helps:** AI-driven procedural difficulty scaling keeps them in a state of optimal challenge ("flow"), while non-punitive Socratic hints rebuild their confidence when answering incorrectly.

## Secondary Audience: Educators & Parents

- **Who They Are:** Teachers and parents seeking verifiable, data-driven insight into student mastery outside the classroom.
- **Core Needs:** Granular analytics showing student solve speed, topic accuracy under time pressure, and specific conceptual roadblocks.
- **Frustrations:** Standard report cards only show final test percentages, failing to reveal *why* a student struggles (e.g., untimed accuracy vs. timed anxiety).
- **Why They Use It:** The integrated Data Visualization (DV) Dashboard translates raw gameplay logs into clear, actionable visual reports on topic mastery and anxiety drop-off points.
- **How It Helps:** Provides real-time evidence of student growth, allowing supervisors to target exact mathematical concepts needing intervention.

# 4. Core Features

## Feature Categorization (MVP Matrix)

### 1. MUST HAVE (Minimum Viable Product)

- **User Profile & Wizard Progression:** A persistent player profile tracking character level, current wizard rank, total experience points (XP), and math accuracy history across tower runs.
- **Procedural Tower Floor Generator:** A floor-by-floor dungeon layout system that procedurally generates varied room encounters (Combat Battles, Mystery Puzzles, and Trap Disarming matching the selected floor difficulty.
- **AI-Driven Core Arithmetic Engine:** Groq API-powered problem generator that dynamically creates endless equations based on core arithmetic: **Addition , Subtraction ), Multiplication , Division , algebra, ect** .
- **Spellcasting Combat Mechanics:** A turn-based battle system where solving arithmetic equations calculates spell power, casting offensive elemental attacks or defensive shields against subterranean beasts.
- **Riddle & Logic Environmental Puzzles:** Non-combat room encounters featuring number-sequence riddles, magic circle algebra puzzles, and chest-unlocking logic equations tied to room progression.
- **South African CAPS Curriculum Mapping:** Baseline mathematical difficulty, topic breakdown, and grade metrics directly aligned with the South African CAPS (Curriculum Assessment Policy Statements) standards for Grades 1–11.

**2. NICE TO HAVE**

- **1v1 Speed Dueling Arena:** A competitive multiplayer or simulated AI duel mode where two wizard apprentices race in real time to answer fast-paced arithmetic sums to overpower each other's magic shields.
- **Expanded Math Curriculum Modules:** Support for advanced topic domains including Fractions, Percentages, Intro to Algebra, Geometry, and Word Problems.
- **International School System Toggles:** Settings option to toggle math standards between SA CAPS, US Common Core, and UK GCSE curriculums.
- **Edge-TTS Voice Mentoring:** Pre-recorded and generated audio triggers using Microsoft Edge-TTS for Archmage voice-over guidance during boss rooms and Socratic hints.

**3. FUTURE**

- Nice to have missed or not done in the time frame.

- **Guilds & Classroom Leaderboards:** Multi-student guilds allowing educators to set custom tower challenges and view class-wide performance metrics.
- **Multi-Class Magic Specialisations:** Pyromancy (Speed Arithmetic), Geomancy (Geometry/Area), and Hydromancy (Fractions/Ratios) skill trees.

## 5. AI Integration

### AI Service & Model

- **Technology:** Groq API running **Llama 3 8B** (Free Tier).

- **Selection Rationale:** Groq provides fast inference speeds necessary for real-time turn-based combat and split-second QTE events, fully accessible within a zero-cost free API tier. I also have some experience in working with it.

## What the AI Does

1. **Procedural Math Problem Generation:** Generates non-repetitive arithmetic equations tailored to the floor level and South African CAPS grade requirements (Grades 1–11).
2. **Full Solution Generation:** creates exact step-by-step worked solutions for every generated sum to validate student inputs in real time.
3. **Targeted Misconception Analysis:** Analyses incorrect student answers mid-battle and provides a 1-to-2 sentence Socratic explanation identifying the exact error (e.g., *borrowing error*, *multiplication mistake*, or *order of operations fault*).

## Why the Application Needs AI

Traditional educational software relies on static, hardcoded question banks that lead to answer memorization and quick user churn. The AI ensures **infinite replayability** and provides an **adaptive mentor system** that guides students through mistakes without breaking gameplay flow.

## What Happens If the AI Feature Is Removed?

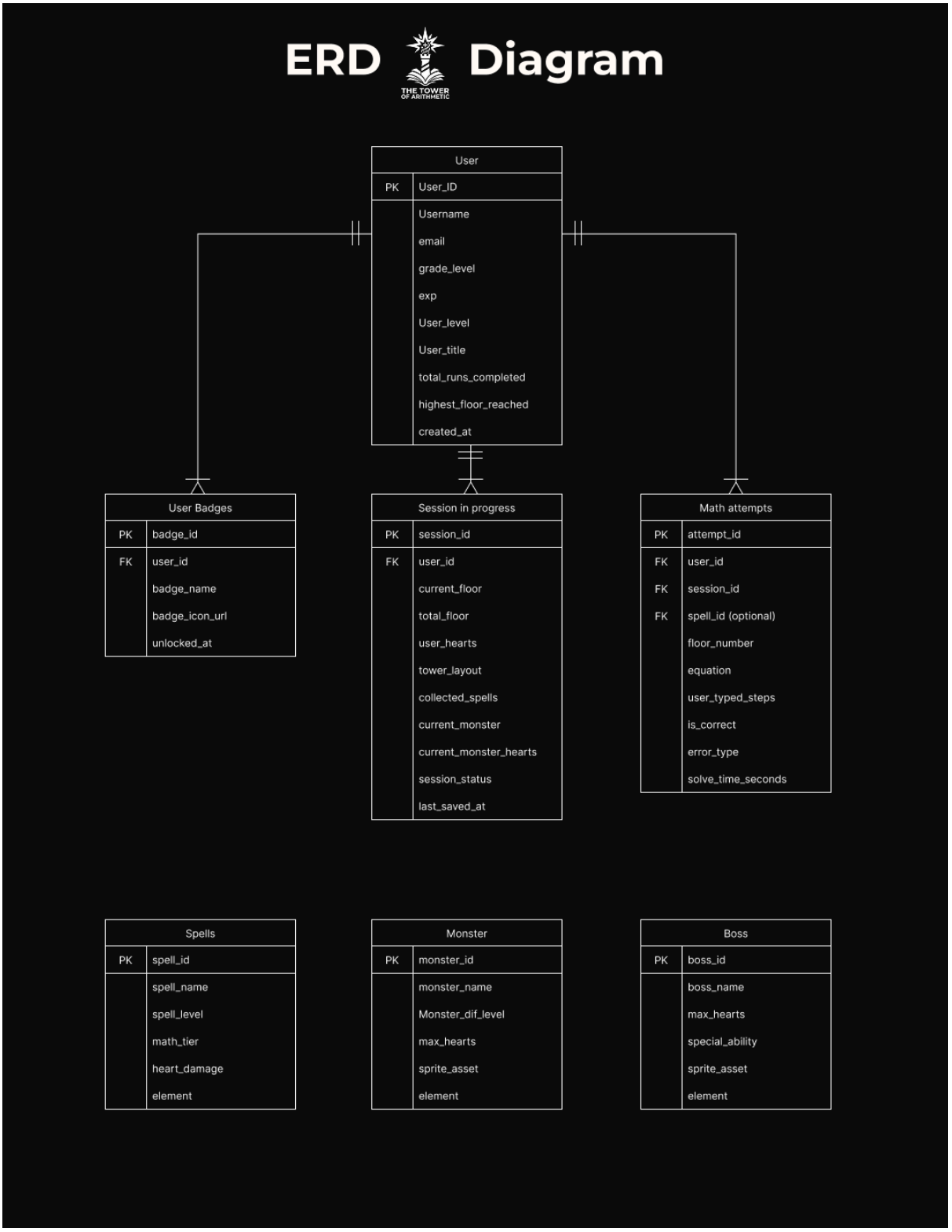
If the AI is removed, *Tower of Arithmetic* breaks fundamentally:

- **Loss of Adaptive Remediation:** The game reverts to a binary "Correct / Incorrect" screen with no explanation, losing its core pedagogical value in reducing math anxiety.
- **Static Question Repetition:** Content becomes finite, predictable, and prone to rote memorization, stripping away the procedural RPG experience.
- **Destruction of Combat Flow:** Without sub-second AI solution generation, real-time QTEs and dynamic spell calculations cannot function.

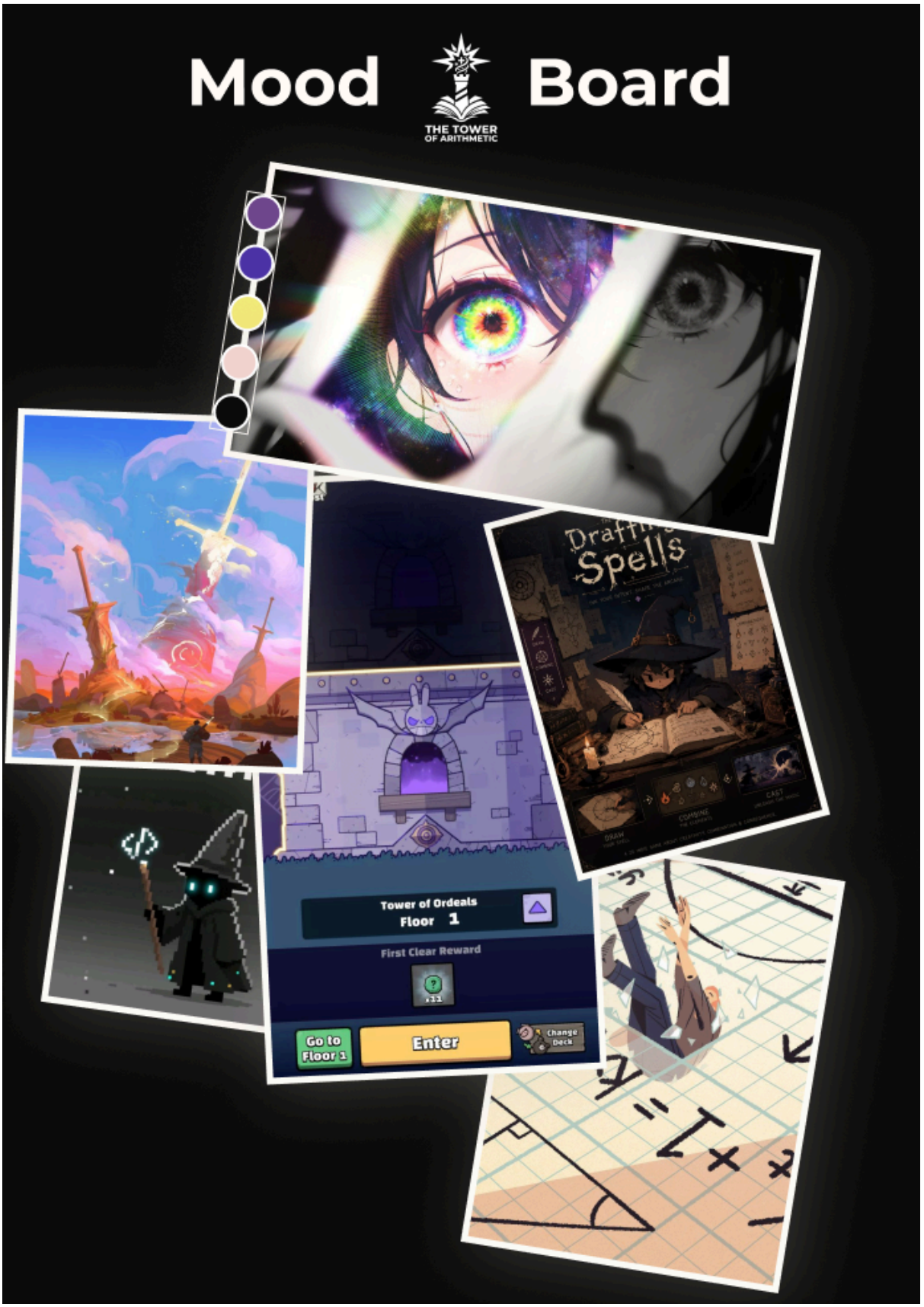
## 6. Technical Planning

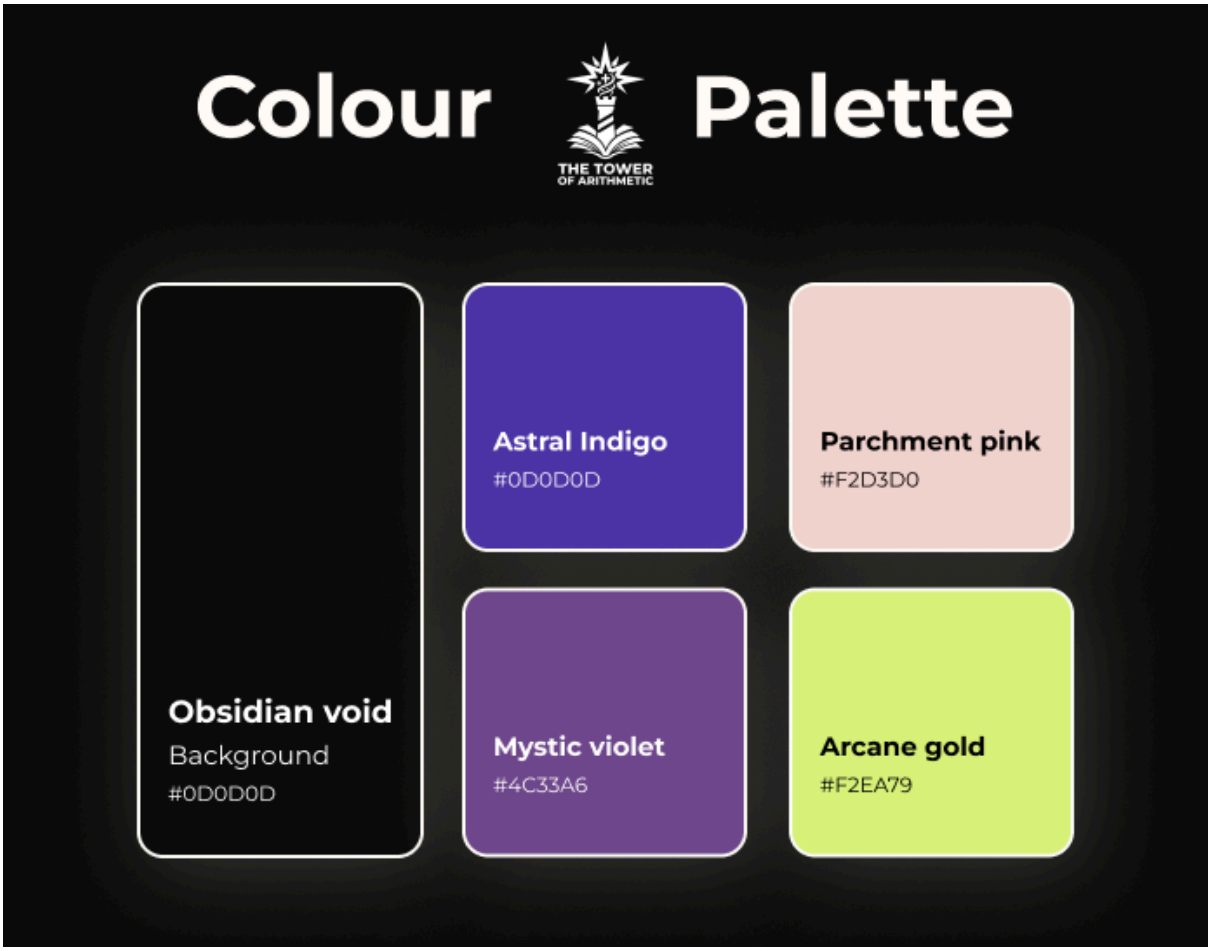
- **Frontend (Mobile App): React Native + Expo (Expo Go)**
    - *Why:* Enables rapid cross-platform mobile development during testing, smooth native UI performance for turn-based combat, and seamless export to a downloadable Android APK.
- **Backend Server: Node.js + Express.js (Hosted on Render)**
    - *Why:* Render provides a reliable, free-tier hosting platform for lightweight Node.js web services. The backend handles secure proxying of the Groq AI API keys (so keys are never exposed on the mobile client) and processes data logic for the app.
- **Database & Auth: Supabase (PostgreSQL)**
    - *Why Supabase over Firebase:* While Firebase is popular for mobile apps, **Supabase is significantly better for this project**. Supabase uses relational PostgreSQL, which makes querying complex aggregated data for your **Educator Data Visualization Dashboard** (e.g., average solve speed, topic accuracy percentages, and progress trends) simple and fast compared to Firebase's non-relational Firestore. It also includes built-in User Authentication.
- **AI Integration: Groq API (Llama 3 8B)** via Node.js backend.

**ERD**

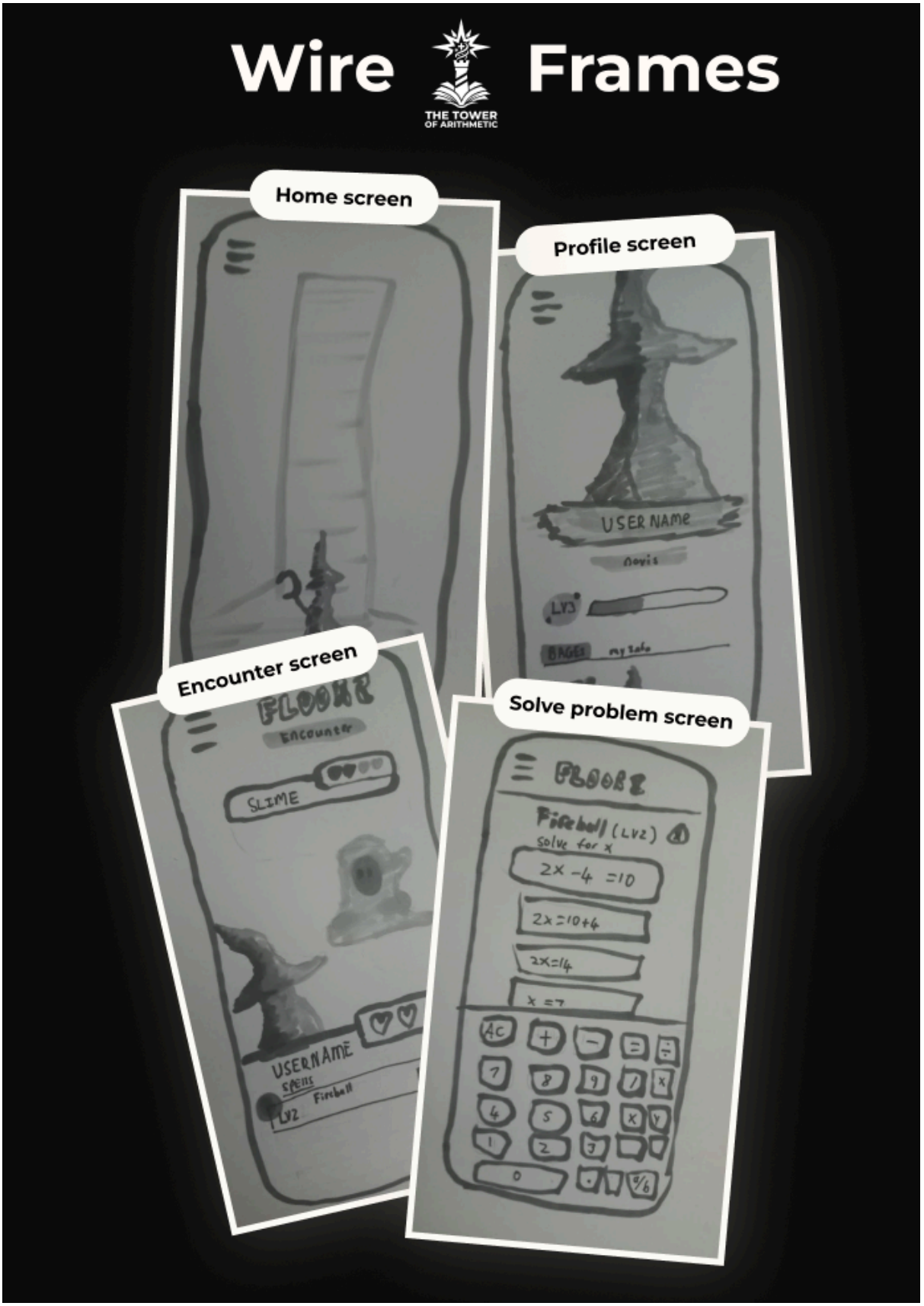

ERD Diagram
THE TOWER OF ARITHMETIC
User
PK User_ID
Username
email
grade_level
exp
User_level
User_title
total_runs_completed
highest_floor_reached
created_at
User Badges
PK badge_id
FK user_id
badge_name
badge_icon_url
unlocked_at
Session in progress
PK session_id
FK user_id
current_floor
total_floor
user_hearts
tower_layout
collected_spells
current_monster
current_monster_hearts
session_status
last_saved_at
Math attempts
PK attempt_id
FK user_id
FK session_id
FK spell_id (optional)
floor_number
equation
user_typed_steps
is_correct
error_type
solve_time_seconds
Spells
PK spell_id
spell_name
spell_level
math_tier
heart_damage
element
Monster
PK monster_id
monster_name
Monster_dif_level
max_hearts
sprite_asset
element
Boss
PK boss_id
boss_name
max_hearts
special_ability
sprite_asset
element

## 7. Brand & UI/UX Direction




Colour Palette
THE TOWER OF ARITHMETIC
Obsidian void
Background
#0D0D0D
Astral Indigo
#0D0D0D
Parchment pink
#F2D3D0
Mystic violet
#4C33A6
Arcane gold
#F2EA79

## 8. Wireframes



## 9. Project Roadmap

### Week 10: Research & Setup

- Finalize UI wireframe flows and color palette.
- Initialize React Native (Expo) app and Node.js backend repositories.
- Write and test Groq AI system prompts for SA CAPS math generation.

### Week 11: Backend & Database Building

- Set up Supabase PostgreSQL tables.
- Implement Supabase authentication for player account logins.
- Deploy Node.js backend service on Render to proxy Groq API keys securely.

### Week 12: AI Engine & Asset Creation

- Connect Groq API (Llama 3 8B) for multi-line algebra validation and Socratic hints.
- Produce sprite artwork and UI icons (wizard, slime monster, spell effects, badges).
- Add Microsoft Edge-TTS voice audio triggers for Archmage hints.

### Week 13: Frontend Development

- Build core screens (Home, Profile, Encounter HUD, Multi-Step Casting).
- Construct a custom 5-column algebra touch keypad.
- Set up screen navigation and state management across views.

### Week 14: Integration & Usability Testing

- Connect React Native screens to backend REST APIs and Supabase.
- Assemble complete turn-based spell combat loop.
- Conduct usability testing with target learners to refine keypad and AI hint mechanics.

### Week 15: Polish, Deployment & Presentation

- Fix edge-case calculation bugs and optimize UI layouts.
- Build downloadable Android APK using Expo EAS.
- Prepare final pitch deck slides and record live demonstration video.

## 10. Conclusion

Math anxiety prevents millions of K–11 students from mastering foundational arithmetic, creating early academic avoidance that hinders long-term STEM readiness. *Tower of Arithmetic* directly addresses this by intrinsically embedding math equations into RPG spellcasting mechanics rather than hiding practice behind abrupt pop-up quiz tollgates. Integrated Groq AI is critical to this experience, serving as an adaptive mentor that generates endless CAPS-aligned equations and mid-combat Socratic hints rather than

punishing mistakes with score drops. Primary and secondary students benefit from a supportive, flow-state learning loop, while parents and educators gain transparent visibility into student growth through granular backend telemetry. Leveraging a proven, lightweight stack of React Native, Supabase, and Node.js, this project is structurally achievable within a six-week sprint—delivering a transformative tool that turns a dreaded academic chore into an empowering fantasy adventure and making *Tower of Arithmetic* a project truly worth building.

## References


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